

service, it solves a big roadblock to cellular adoption. The language-agnostic JSON based programming API enables an unmatched developer experience. Available in just 30mm × 35mm size, the hardware is power conscious and draws very little current (~8µA when idle) for secure 'off the internet' communications from device-to-cloud.

Blues Wireless
www.blues.io

4D imaging radar

The 79GHz AIR 4D is an imaging radar solution featuring multi-chip cascading technology to improve target detection, produce elite high-resolution 4D imaging, and aid in autonomous driving



collision avoidance to drastically reduce crashes and fatalities occurring on the roads today. The scalable data from AI machine-learning allows for AIR 4D to precisely understand surroundings and classify target objects solely through radar perception using the integrated camera sensor. From design to mass production, the AIR 4D is capable of transforming autonomous driving and improving the way all vehicles understand surroundings.

bitsensing
www.bitsensing.com

Processor for AI at the edge

The FPGA based spiking neural network (SNN) accelerator, known as Akida, comprises 80 neuromorphic processing units (NPU), providing 1.2 million virtual neurons and 10 billion virtual synapses. Each NPU comes with 100kB of SRAM memory and is networked into an on-chip mesh. It also has the capability to run convolutional neural networks (CNNs). AI application areas include embedded vision, embedded

audio, automated driving (lidar, radar), cybersecurity, and industrial IoT. Keeping with the low-power computation demands of edge devices, it uses 1-bit,



2-bit, or 4-bit precision for each CNN layer, instead of the typical 8-bit precision, saving both energy and memory space.

BrainChip
www.brainchipinc.com

Wireless temperature, humidity, and motion sensor

The CubiSens AH110 is a wireless, small-size (similar to a vitamin pill) programmable smart tag ideal for unobtrusive



wireless monitoring of movement, temperature, and humidity. It can transmit data and send an alarm to the gateway up to 200 metres away while running for up to seven years. The CubiSens AH110 can be used for securely monitoring everything from a door opening to the operating temperature of a cooler, without wires or supplemental power. The applications for environmental and security monitoring are endless.

CubeWorks
www.cubeworks.io

Tiny MEMS speaker for TWS and hearing-aids

At just 22mm³, 56 milligrams and 3.4mm diameter side-firing package, Cowell provides up to 15dB gain above 1kHz for improved speech-in-noise performance

monolithic MEMS µspeaker that delivers an impressive 110dB SPL at 1kHz. Cowell provides up to 15dB gain above 1kHz for improved speech-in-noise performance



and greater vocal and instrumental clarity versus electrodynamic and balanced armature µspeakers. Cowell uses xMEMS' 2 proprietary 2nd generation M2 speaker cell architecture offering improved SPL/mm² for increased loudness in smaller form factors.

xMEMS
www.xmems.com

Depth camera for robotic tracking and navigation

The G53 is a stereo depth camera with 60 degrees of FOV for robotics depth capture and object tracking. The camera includes two mono sensor pairs for



various resolutions of stereo, mono, and depth disparity/distance map output via USB. Global shutter mono sensors and IR dot projector make the product ideal for robots, AGV/AMR, G2P as well as fast-motion depth capturing development. It outputs a clean depth map and requires minimum host computing support that is suitable for wide range of depth applications in fast-moving systems with excellent indoor and outdoor depth output performance.

eCapture
www.ecapturecamera.com